
Hands-On Lab: Connect EC2 Using AWS Systems Manager (Session Manager)

Objective

Use **AWS Systems Manager Session Manager** to securely connect to an **EC2 instance** via the web console — **no SSH, no open ports, no key pair required**.

Prerequisites

Requirement	Details
AWS Account	With access to EC2 & Systems Manager
IAM Permissions	<code>AmazonSSMFullAccess</code> (for user)
Region	Any region (e.g., <code>us-east-1</code>)
VPC	Public or private subnet is OK (no Internet needed)

What You'll Do

-  Launch an EC2 instance
 -  Attach the correct IAM role
 -  Connect to EC2 using **Session Manager** in the AWS Console
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Step 1: Create an IAM Role for SSM

1. Go to **IAM Console** → Roles → **Create role**
 2. Select **Trusted Entity Type**: AWS Service
 3. Use Case: **EC2**
 4. Click **Next**, then attach this policy:
 - `AmazonSSMManagedInstanceCore`
 5. Name the role: `EC2SSMRole`
 6. Click **Create role**
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Step 2: Launch EC2 with SSM Role

1. Go to **EC2 Console** → Instances → **Launch instance**
 2. Name: `SSM-Test-Instance`
 3. AMI: **Amazon Linux 2** (SSM agent pre-installed)
 4. Instance type: `t2.micro` (Free Tier)
 5. Key pair: **None required**
 6. Network: Choose default VPC
 7. IAM role: **Select** `EC2SSMRole`
 8. Storage: Keep default (8 GB)
 9. Click **Launch**
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Step 3: Verify Instance is Ready for SSM

1. Wait until **instance state = running**
2. Go to **Systems Manager** → **Fleet Manager**
3. If properly configured, your EC2 instance will appear as **Managed**

✅ If it doesn't show:

- Ensure SSM Agent is running
 - Ensure IAM role has correct policy
 - Check instance is in **running** state
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Step 4: Connect Using Session Manager

1. Go to **Systems Manager** → **Session Manager**
2. Click **Start session**
3. Select your EC2 instance
4. Click **Start session**



You now have shell access to the EC2 instance — **without SSH or a key pair!**



Optional: Run Commands via Run Command

1. Go to **Systems Manager** → **Run Command**
2. Click **Run a command**
3. Choose **AWS-RunShellScript**
4. Target: Select the instance

Command:

uptime

- 5.
6. Click **Run**
7. View output after execution

Cleanup

To avoid charges:

- Stop or terminate the EC2 instance
- (Optional) Delete the IAM role `EC2SSMRole`

Summary

Task	Completed
Created IAM role for SSM	
Launched EC2 with IAM role	
Connected via Session Manager	
Ran remote shell command	
